

Table 1: Various Nmap scripts

No.	Script	Function
1	<i>dhcp-discover</i>	Discovers DHCP servers on the network; the UDP <i>discover</i> request is sent from port 67, and the response is received on port 68.
2	<i>ftp-bounce</i>	Checks whether there are FTP servers that allow an FTP bounce attack to other hosts on the network. (Please refer to earlier articles, where we have discussed FTP bounce attacks.)
3	<i>http-iis-webdav-vuln</i>	Checks whether hosts with vulnerabilities listed in Microsoft security bulletin MS09-020 (IIS 5.1/IIS 6.0) are present on the network.
4	<i>ms-sql-info</i>	Identifies Microsoft SQL Server details.
5	<i>mysql-info</i>	Identifies MySQL Server details.
6	<i>p2p-conficker</i>	Checks whether a host is infected by the conficker.c worm.
7	<i>smb-enum-shares</i>	A very interesting script, which identifies all SMB shares within the specified address range.
8	<i>smb-enum-users</i>	Will identify all SMB user names within the specified address range.
12	<i>sniffer-detect</i>	Finds hosts with pcap libraries installed.
13	<i>sshv1</i>	Lists all hosts with SSH version 1. As most of you will be aware, this version has documented vulnerabilities.

The various command line options for script scanning are as follows:

—sC: basic script scan.

—script-updatedb: will update the script database.

Some of the scripts may ask for the —script-args=unsafe option to be set to 1. Please remember, this is a destructive test; these scripts will almost surely crash a system with the corresponding vulnerability. Be careful you do not use this option in a live production environment.

The actual scans

To start with, create a file *IPList.txt*, listing all the active hosts on the network, to specify target hosts to NMap. Next, try to analyse the results of a few scans performed in a live environment. When you wish to try these scans, don't forget to take written permission from the management.

Checking for SSHv1 (SSH - Secured Shell), v1 - Version 1

Table 2: SSHv1 scans

Command/details	Explanation
<i>nmap—script sshv1 -iL IPList.txt -osshv1.txt</i>	The command line, running only one script, <i>sshv1</i> .
Masked output of <i>sshv1.txt</i> :	
<i>Nmap scan report for 192.168.1.4</i> <i>Host is up (0.00011s latency).</i> <i>Not shown: 984 closed ports</i> <i>PORT STATE SERVICE</i> <i>21/tcp open ftp</i> <i>22/tcp open ssh</i> <i> _sshv1: Server supports SSHv1</i> <i>80/tcp open http</i> <i>111/tcp open rpcbind</i> <i>113/tcp open auth</i>	The output identifies that a host with IP address 192.168.1.4 is running SSHv1. On further probing, the host identified it to be running a very old version of Red Hat Linux. Recommendation: Upgrade SSHv1 to the latest version.

Sniffer detection

Table 3: SnifferDetectScan

Command/details	Explanation
<i>nmap—script sniffer-detect -iL IPList.txt -osniffer-detect.txt</i>	Initiating sniffer detection
Masked output of <i>sniffer-detect.txt</i>	
<i>Nmap scan report for 192.168.1.26</i> <i>Host is up (0.00012s latency).</i> <i>Not shown: 992 closed ports</i> <i>PORT STATE SERVICE</i> <i>135/tcp open msrpc</i> <i>139/tcp open netbios-ssn</i> <i>445/tcp open microsoft-ds</i> <i>2967/tcp open symantec-av</i> <i>5101/tcp open admdog</i> <i>5225/tcp open unknown</i> <i>5226/tcp open unknown</i> <i>8008/tcp open http</i> <i>MAC Address: 00:XX:XX:XX:XX:2D (Intel Corporate)</i> <i>Host script results:</i> <i> _sniffer-detect: Windows with libpcap installed; may or may not be sniffing (tests: "1_1__1_")</i>	The detection of host 192.168.1.26 informs you that it is running Windows, with <i>libpcap</i> installed. It further states that the sniffer may not be sniffing. After discussion with the management, it became clear that for testing their environment, pcap libraries were indeed installed on this host.